Radon Monitoring

User manual

Local monitoring for GQ RadonScan devices

Version	5.5.14
Date	13 August 2026
Home Assistant	Local app / Ingress
Data	Local SQLite + optional external upload

Contents

Important: this app is a monitoring and documentation tool, not an official measurement system.

1. Purpose, measurement principle and limitations

Radon Monitoring reads completed hourly records from compatible GQ RadonScan devices using read-only SPIR commands. It stores raw counts (CPH), the conversion factor used for each record and the calculated activity concentration locally in SQLite.

The app supports continuous observation, statistical analysis and traceable documentation. It does not replace traceable calibration, a recognised regulatory measurement or professional medical, building or legal advice.

A high individual hourly value is not automatically an exceedance of an annual reference value. The interface and reports distinguish the current value, period statistics, counting uncertainty, calibration information and professional interpretation.

2. Installation, upgrade and first start

Connect the RadonScan by USB, make MQTT available to Home Assistant and start the app. Version 5.5.14 defaults to /dev/serial/by-id/usb-1a86_USB_Serial-if00-port0. A fixed serial_port is used exclusively; auto deliberately enables discovery.

Create a Home Assistant backup before upgrades and destructive actions. After upgrading, the sidebar or Help view must show version 5.5.14. If an old Ingress view remains open, close the panel and reopen it.

Only completed hours are imported. A newly connected device can therefore remain without a current value until a complete record is available.

3. Overview

The Overview context bar selects a device and campaign. The most recently or currently assigned room is resolved automatically. Current value, period windows, selected sample count, 24-hour maximum and chart are derived from that device, campaign and room. Device and campaign choices are retained locally in the browser.

The Radon traffic light primarily assesses the 24-hour mean. Until duration or coverage is sufficient, it provisionally uses the latest completed hourly value. The assessment basis, coverage and configured warning/danger thresholds are displayed directly below the signal.

The fourth metric shows the observed maximum within the latest 24-hour window, including its time and coverage, instead of a potentially mixed overall mean. Documented events appear as chart markers and gaps are shown as breaks rather than connected lines.

Depending on the privacy mode, the Home Assistant location card shows an address and precise coordinates, only the location name and rounded coordinates, or no location card. Coordinates include degree units and cardinal directions, for example 51.60176° N, 7.45410° E. Directly below it, the interface shows only the automatically assigned room and, when available, the measurement height. Location values come only from the local Home Assistant Core configuration and no external geocoding is performed.

4. Analysis and statistics

Analysis starts with a plain-language summary of trend direction, warning/danger threshold share and data coverage. Basic analysis contains mean, median, minimum, maximum, quantiles and threshold time. Effective sample size, moving-block-bootstrap confidence intervals, autocorrelation, distribution diagnostics and sensitivity analysis are grouped in an advanced section that is collapsed by default.

Missing measurements are not imputed. Unusual observations are not deleted automatically. Quality flags identify undocumented factors, non-primary sources, implausible values and duplicate timestamps.

Change-point analysis is exploratory. For very long series, only this diagnostic is bounded to an evenly distributed, time-ordered sample. Descriptive statistics, threshold calculations and data checksums continue to use every selected record.

Statistical significance proves neither causality nor practical importance. Events and interventions still require professional interpretation.

5. Devices & System, factor and calibration

The former Expert view was removed completely from the interface and sidebar in version 5.1.0. Device, protocol, service and database status is grouped under Devices & System; configured thresholds and other operating parameters are shown under Settings.

Home Assistant receives only three MQTT sensors: the completed hourly value, the 24-hour mean and the 7-day mean. All three always use Bq/m^3, independently of the display unit selected inside the app. Previous diagnostic entities and the 30-day sensor are removed from discovery after the first successful MQTT connection following the update. The three requested sensors are immediately republished on a stable RadonScan device identifier, even if the USB device is not connected yet.

The on-demand System self-test checks the API version, SQLite integrity, a rolled-back room persistence round trip, the report directory, device runtime, MQTT and the local Home Assistant connection. Results are shown as an actionable checklist and do not alter stored rooms or measurements.

The currently configured factor is displayed separately from the factor stored with the selected historical measurement. Historical results therefore remain reproducible after later configuration changes.

Calibration records can include date, laboratory, certificate reference, factor, uncertainty, next due date and notes. A calibration entry does not silently alter existing measurements.

6. Rooms, assignments and events

The Local metadata view is named Rooms and events. Place or address and building name are taken directly from the general Home Assistant settings and cannot be entered again or differently in the app. Only the room and measurement height are added manually.

The workflow is divided into two steps: first create or edit a room, then assign it to a device and time period. Assignment remains disabled until at least one room has been stored. Validation messages appear directly beside the affected field, and stored assignments are shown in a dedicated list below the forms.

Room storage is idempotent: saving the same normalised room name again updates the existing record instead of creating a duplicate. Both decimal points and decimal commas are accepted for measurement height, and under Home Assistant Ingress the interface sends tightly bounded fallback information if a proxy changes the regular request body.

Measurements can be assigned to a room for a defined period. Campaign, purpose, start, end and notes remain local. Ventilation, interventions, device moves, construction work or outages can be documented as events without changing original measurements. Matching events are marked in Overview and Analysis charts.

7. GQ Radiation World Map

Upload is optional and disabled by default. When gmcmap_enabled is off, the World Map view is omitted from the sidebar. When enabled it contains only external upload, queue and history functions; local rooms, assignments and events remain in their dedicated view. Account ID and device ID are shown only in masked form.

A persistent queue prevents duplicates, retries temporary failures with increasing delay and can discard measurements older than a configured limit. Manual upload is a protected POST action. Upload history shows ten entries per page, with arrows moving backwards or forwards by ten records.

RadonScan values are sent through GMCMap's documented public log2.asp endpoint using only AID, GID and pCi. The app does not submit CPM, ACPM or uSV. New uploads therefore no longer create artificial 0 CPM radioactivity entries.

The public position is managed in the GQ account. Radon Monitoring does not transmit its own GPS coordinates. Zero-valued records uploaded previously are stored by the external GMCMap service and are not removed by an application update.

8. Scientific PDF reports

Compact and detailed reports contain period, room, measurement height, Home Assistant location and building, device, factor, coverage, statistics, time series, method, quality information, software version and SHA-256 checksums.

For multi-year series, only the time-series chart is reduced to a bounded peak-preserving point set. Statistics, threshold events and the data checksum continue to use all selected measurements.

Reports are documentation aids, not official certificates.

9. Data management and security

The complete Data management area can be hidden from navigation and blocked at the API using data_management_enabled.

Complete local reset automatically creates a safety backup, clears app data transactionally, verifies SQLite integrity and reports operation ID, deleted records, removed files, sizes and duration. Immediate re-import of the deleted device history is suppressed.

Home Assistant Recorder purge submits recognised RadonScan entities and stable name patterns to recorder.purge_entities. An optional long-lived administrator token may be required. It is never displayed, excluded from diagnostics and centrally redacted from error messages.

Verification subsequently checks recent Home Assistant history. It does not prove immediate physical compaction of every Recorder backend and does not automatically cover separately retained long-term statistics. The former Administration log with raw JSON details has been removed from the interface; internal audit entries remain stored for traceability and diagnostics.

10. Operation, mobile layout and accessibility

On small screens the sidebar opens as a menu. Tables and the weekly heatmap remain horizontally scrollable inside their own cards; the rest of the page must not create horizontal page overflow. Hourly history and World Map uploads show ten entries per page and use arrow buttons to move in groups of ten.

The interface supports keyboard operation, visible focus, a skip link, Escape to close the menu, reduced motion and text scaling to 200 percent.

Version 5.5.14 additionally supports long-running RadonScan histories whose visible time window no longer begins with the original t=0 marker. Such a history is accepted only when all visible time records remain exactly hour-aligned and continuous at 3600-second intervals. Stored values and the database schema remain unchanged.

11. Troubleshooting

No device: check USB mapping, permissions, configured port and competing processes.

No new values: only completed hours are imported. Press Refresh and inspect the latest scan time under Devices & System.

Old interface: restart the app, close the Ingress panel and reopen it.

Recorder purge fails: check administrator token, Recorder integration and Home Assistant administrator rights.

Verification reports remaining data: wait and verify again; Recorder maintenance can be asynchronous. Treat long-term statistics separately.

Restore rejected: only readable SQLite backups with a compatible schema are accepted. The active database remains unchanged after validation failure.

12. Privacy and responsible use

Measurements and metadata remain local by default. Only enabled external functions transmit data. Location, address and building name are read only through the local Home Assistant Core API and are not manually duplicated. No external geocoding service is used. location_display_mode can show full, reduced or no location details. Before screenshots, reports and World Map publication, confirm that visible location data meets your privacy requirements.

Protect backups and reports because they may contain device identifiers, rooms, periods and building information.

13. New in version 5.5.14

Version 5.5.14 fixes the repeated first time record is not t=0 error on long-running RadonScan devices. Fresh histories keep the existing interpretation of t=0 as a marker without its own CPH value. If the visible time window instead begins at a later hour, it is accepted as a rolling history only when every time record is hour-aligned and the sequence advances by exactly 3600 seconds. In that state each visible time record receives one raw CPH value. Block-length, sequence and plausibility checks remain active; the database schema and stored measurements are unchanged.

Configuration overview

Option	Purpose
serial_port	Fixed RadonScan path; use auto only for deliberate discovery
factor_bq_m3_per_cph	Currently configured conversion factor
minimum_data_coverage_percent	Minimum coverage for period results and quality
analysis_timezone	Time zone for daily and weekly profiles
location_display_mode	Show full, reduced or no location details
data_management_enabled	Enable Data management navigation and API
homeassistant_access_token	Optional administrator token for Recorder actions
gmcmap_enabled	Enable GQ Radiation World Map
gmcmap_auto_upload	Enable the automatic upload queue

Further technical details are available in DOCS.md and the built-in Help view.